

SE Lab

(UE24CS341A)

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1. Problem Statement & Context

Problem: Blood banks require a real-time stock management system that monitors blood component shelf lives and triggers geo-targeted emergency notifications to matching eligible donors during critical shortages.

Target stakeholders / actors identified in the problem statement: Emergency Requester and Blood Bank Manager. The use-case model additionally represents the Donor and the external SMS Gateway involved in the described workflow.

2. Requirements Table

Five Functional Requirements (FR-001 to FR-005) and two Nonfunctional Requirements (NFR-001 to NFR-002) are included.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-1	Functional	The system shall cross-reference emergency blood unit requests against real-time blood bank inventory and broadcast SMS alerts to compatible donors within a 10 km radius.	High	Pass: Compatible donors notified within 30 seconds of emergency flag. Fail: Incompatible blood group alerted.	Given requirement (baseline). Ensures the core emergency-matching function works correctly and quickly.
FR-2	Functional	The system shall verify ABO and Rh(D) compatibility between the requested blood type and each candidate donor's blood type before adding that donor to the notification list.	High	Pass: In test runs covering all 8 blood group combinations, zero incompatible donors appear in the notification list. Fail: Any incompatible donor is included.	Prevents life-threatening transfusion errors caused by incorrect blood-type matching.
FR-3	Functional	The system shall allow a Blood Bank Manager to add, remove, or adjust inventory counts by blood group and component type (whole blood, plasma, platelets).	High	Pass: Inventory count reflects the manager's change within 2 seconds and is visible on the dashboard. Fail: Count remains stale after 2 seconds or does not match the entered value.	Accurate, current stock levels are required before the system can correctly decide whether an emergency donor search is needed.
FR-4	Functional	The system shall allow verified medical centres to log high-urgency blood requests, capturing the required quantity, urgency level (Immediate/Code Red vs. Scheduled Surgery), and patient blood profile.	High	Pass: A verified medical centre can submit a request that captures quantity, urgency level, and blood profile, and the system rejects submission if the requester is unverified or any required field is missing. Fail: A request is accepted without an urgency level/blood profile, or from an unverified medical centre.	Ensures accurate triage so that Immediate/Code Red requests are prioritised for matching and fulfillment ahead of scheduled, non-emergency requests.
FR-5	Functional	The system shall automatically flag blood units nearing expiry (within 48 hours) on the Blood Bank Manager's dashboard.	Medium	Pass: 100% of units with less than 48 hours to expiry are shown flagged (e.g., in red) on the dashboard. Fail: Any qualifying unit is left un-flagged.	Reduces wastage of perishable blood units and supports proactive stock rotation.

NFR-1	Nonfunctional (Performance & Security)	The inventory ledger must maintain strict transactional consistency, preventing simultaneous allocation of the same blood bag.	High	Pass: Benchmarking/load tests confirm target latency and no double-allocation under simulated peak concurrent load. Fail: Any test run produces a double-allocated unit or exceeds target latency.	Given requirement (baseline). Prevents double-booking of a single physical blood unit to two different requests.
NFR-2	Nonfunctional (Performance & Security)	The system shall encrypt all donor personal and medical data (blood group, contact number, location) both in transit and at rest.	High	Pass: Security audit confirms TLS 1.2+ is used for all data in transit and AES-256 for all data at rest, with no field found in plaintext. Fail: Any donor data field is found unencrypted in transit or storage.	Protects sensitive personal health information and supports compliance with data-protection regulations.

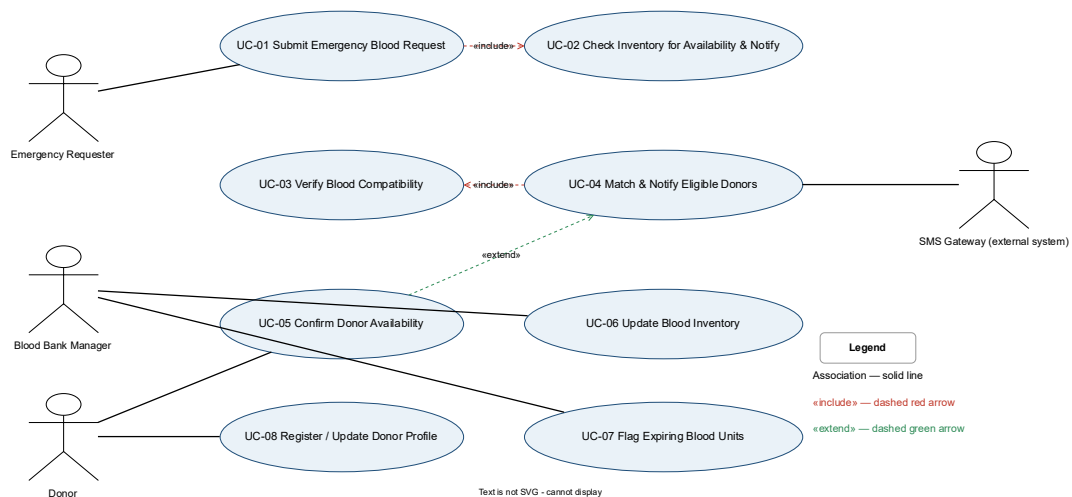
3. Actors and Use Cases

Actor	Role / Interaction
Emergency Requester	Submits a high-urgency blood request with quantity, urgency level, and patient blood profile.
Blood Bank Manager	Maintains inventory and views/acts on expiring blood units.
Donor	Provides/updates donor profile and confirms availability after notification.
SMS Gateway (external system)	Delivers SMS notifications to eligible donors.

Use Case ID	Use Case
UC – 01	Submit Emergency Blood Request
UC – 02	Check Inventory for Availability & Notify
UC – 03	Verify Blood Compatibility
UC – 04	Match & Notify Eligible Donors
UC – 05	Confirm Donor Availability
UC – 05	Update Blood Inventory
UC – 06	Flag Expiring Blood Units
UC – 07	Register / Update Donor Profile

Key UML relationships: UC-01 «include» UC-02; UC-04 «include» UC-03; UC-05 «extend» UC-04. These satisfy the handout's requirement for include/extend relationships.

4. UML Use-Case Diagram



5. Use-Case Flow Specification

Selected Use Case: UC-01 — Submit Emergency Blood Request

Goal: Allow a verified medical centre / emergency requester to submit a complete high-urgency blood request so the system can check availability and initiate donor matching when required.

5.1. Preconditions

- The requester is a verified medical centre / authorized emergency requester.
- The Blood Bank Management System is available and the inventory ledger is accessible.
- The requester has the required blood request information: quantity, urgency level, and patient blood profile.

5.2. Main Success Scenario

1. The Emergency Requester selects the option to submit an emergency blood request.
2. The system displays the blood request form.
3. The requester enters the required quantity, urgency level (Immediate/Code Red or Scheduled Surgery), and patient blood profile.
4. The system validates the requester and checks that all mandatory fields are present.
5. The system records the emergency blood request.
6. The system cross-references the request against real-time blood bank inventory.
7. If available compatible stock is insufficient for the request, the system initiates eligible-donor matching.
8. The system verifies ABO and Rh(D) compatibility for candidate donors before notification.
9. The system sends SMS alerts to compatible donors within the 10 km target radius through the SMS Gateway.
10. The system records the request and notification status and displays confirmation to the requester.

5.3. Alternate Flow-Request Validation / Unverified Request

- The system finds that the requester is not verified, or one or more required fields are missing.
- 4a1. The system rejects the request and displays the missing/invalid information.
- 4a2. The requester corrects the information and resubmits the request.
- 4a3. If the requester cannot be verified, the request is not accepted and no donor notification is initiated.

5.4. Postconditions

- The valid emergency request is stored with its quantity, urgency level, and patient blood profile.
- The system has either identified sufficient inventory or initiated the donor-matching workflow.
- Any donor notification generated by the workflow is associated with the emergency request for tracking.